

Palm kernel shell co-firing: life-cycle results

Brightway 2.5 model · ReCiPe 2016 (H) midpoint proxy · generated 2026-08-20 10:51 UTC

Functional unit. 1 MWh net electricity delivered to the Java-Bali grid.
System boundary. Cradle-to-grave for the fuel supply and conversion service: allocated palm kernel shell (PKS) upstream burden, drying and size reduction, transport (truck and barge), on-site handling, boiler combustion in a 300 MW sub-critical PLTU unit, FGD reagent supply, process water, and controlled ash (FABA) landfilling. Plant construction and decommissioning are excluded (cut-off), consistent with the original ISO 14067 study.

1. Electricity background model in force

Auxiliary and crusher electricity are charged at the **Attributional grid average (time of use, 2025)** factor of 0.794 kg CO₂-eq/kWh (mode key `average_time_of_use`; other indicator columns scaled by 1). Source: KESDM Jamali grid emission factor 2025.

	label	gwp_kg_per_kWh	co_scale	source
mode				
average_time_of_use	Attributional grid average (time of use, 2025)	0.794	1	KESDM Jamali grid emission factor 2025

2. Scenarios

	co-firing (% energy)	truck (km)	barge (km)	net efficiency (%)
scenario				
baseline_0pct	0	100	0	33
cofire_5pct	5	100	0	32.7
cofire_10pct	10	100	0	32.3
cofire_20pct	20	150	0	31.7
cofire_30pct_barge	30	60	350	31

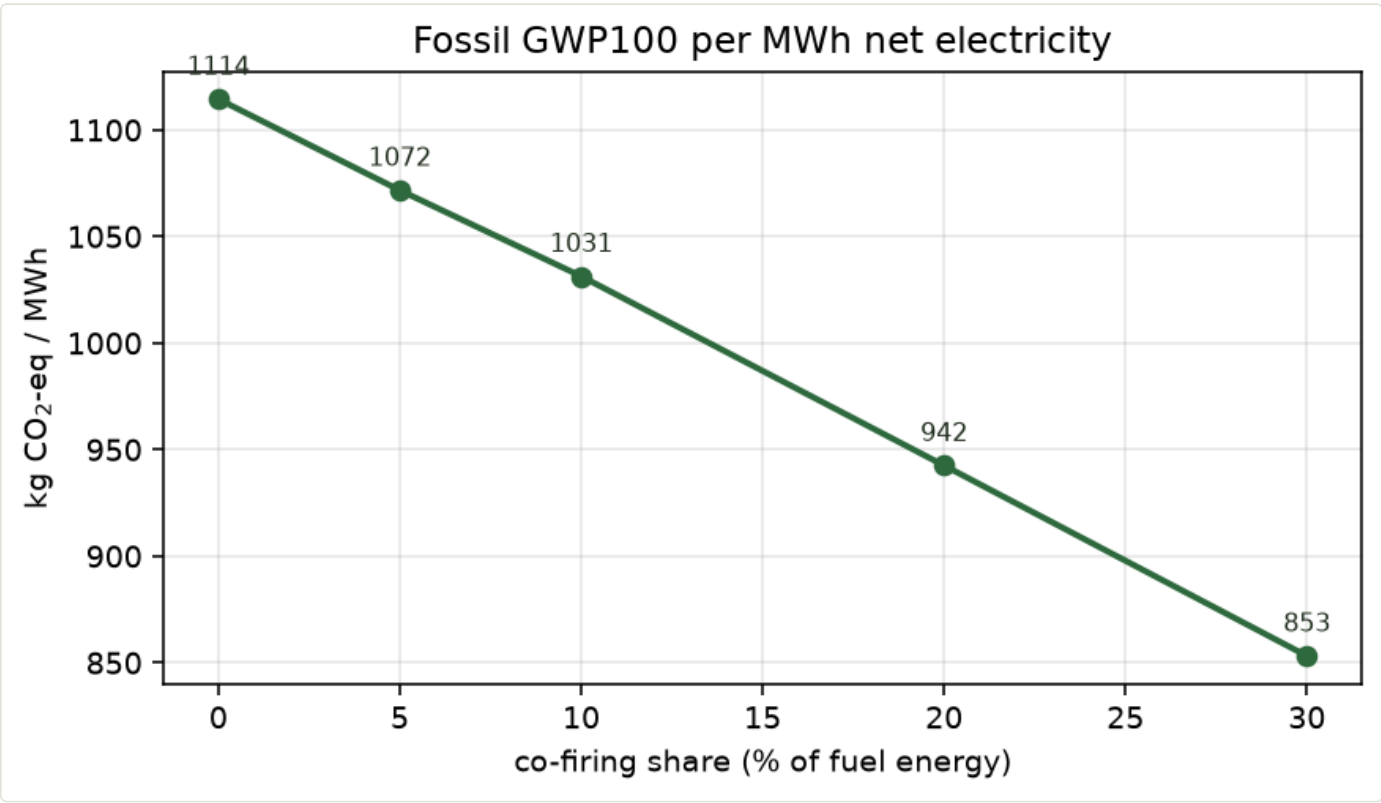
3. Parameters and sources

Every number a reviewer can challenge, with its provenance.

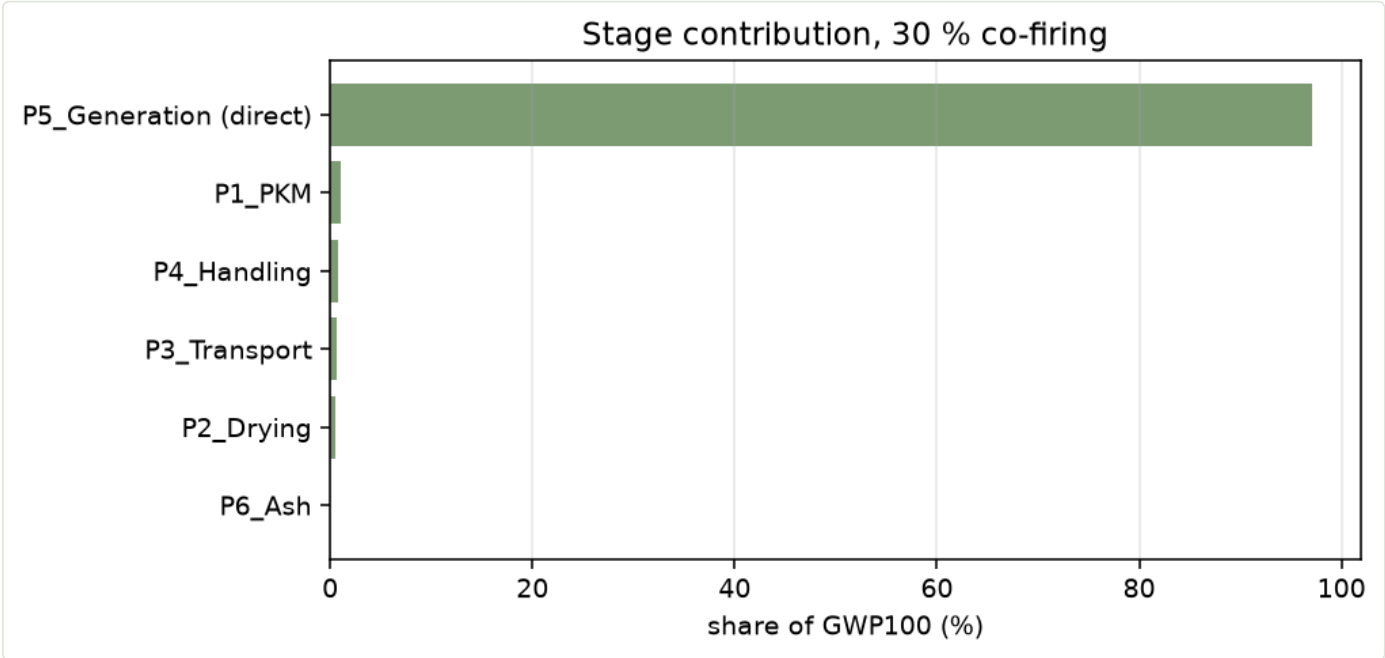
	value	source
parameter		
lhv_coal_MJ_per_kg	18.626	lab mean, sub-bituminous 4200 kcal GAR
lhv_pks_MJ_per_kg	17.035	lab mean, PKS as received
ash_coal_frac	0.055295	lab mean, dry basis
ash_pks_frac	0.030115	lab mean, dry basis
s_coal_frac	0.0040965	lab mean, dry basis
s_pks_frac	0.0004885	lab mean, dry basis
n_coal_frac	0.0095485	lab mean, dry basis
n_pks_frac	0.004059	lab mean, dry basis
co2_coal_kg_per_GJ	94.6	IPCC 2006, sub-bituminous coal
co2_pks_kg_per_GJ	110	IPCC 2006, other primary solid biomass (biogenic)
ch4_kg_per_GJ	0.001	IPCC 2006, 1 g CH4/GJ
n2o_kg_per_GJ	0.0015	IPCC 2006, 1.5 g N2O/GJ
nox_coal_kg_per_GJ	0.1485	CEMS Unit #2 annual mean 2025
so2_coal_kg_per_GJ	0.19708	CEMS + 90 % FGD removal
pm25_coal_kg_per_GJ	0.008067	annual stack test, ESP 99.5 %
ash_capture_frac	0.905	ash balance, B3 landfill weighbridge
limestone_kg_per_kg_S	1.407	FGD reagent store issue records
aux_kWh_per_kg_fuel	0.01204	own-use log, conveyor and coal mill
water_m3_per_MJ_fuel	0.000165	WTP water balance
pks_upstream_kg_per_kg	1	economic allocation 3.1 % embedded in background dataset
drying_heat_MJ_per_kg	0.85	dryer energy balance
crusher_kWh_per_kg	0.01808	crusher nameplate 75 kW and throughput log
stockpile_ch4_kg_per_kg	0.00021973	biomass storage literature
land_m2a_per_kg	0.42	3.8 t CPO/ha/yr productivity, allocated

4. Impact results per functional unit

	GWP100 (climate change)	Terrestrial acidification	Photochemical ozone formation	Fine particulate matter formation	Freshwater eutrophication	Land use	Water consumption
scenario							
baseline_0pct	1114	3.324	1.876	2.153	0.1872	0	1.8
cofire_5pct	1072	3.23	1.848	2.109	0.1858	13.57	1.817
cofire_10pct	1031	3.144	1.825	2.071	0.185	27.48	1.839
cofire_20pct	942.5	2.951	1.77	1.99	0.1839	56	1.874
cofire_30pct_barge	853.1	2.771	1.749	1.938	0.1837	85.9	1.916

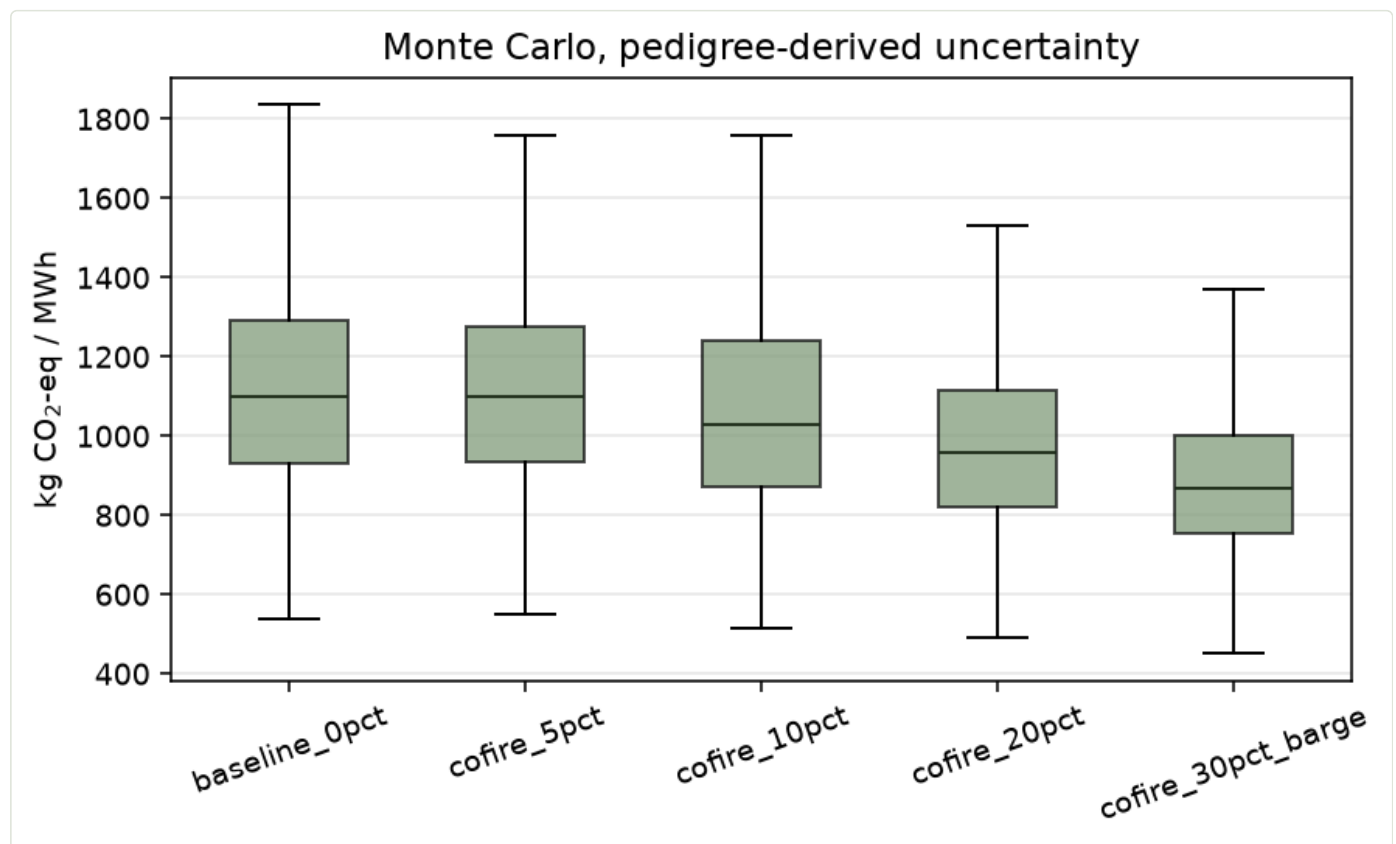


5. Contribution analysis



stage	score	share_pct
P5_Generation (direct)	827.8	97.03
P1_PKM	8.794	1.031
P4_Handling	6.127	0.7182
P3_Transport	5.301	0.6213
P2_Drying	4.879	0.5719
P6_Ash	0.2166	0.02539

6. Monte Carlo under pedigree uncertainty



	mean	median	p5	p95	cv_pct	p_better_than_baseline
scenario						
baseline_0pct	1130	1101	763.8	1573	22.7	0
cofire_5pct	1117	1098	729.7	1581	23.36	0.52
cofire_10pct	1067	1028	682	1570	25.4	0.5775
cofire_20pct	986	958.5	654.8	1385	23.42	0.67
cofire_30pct_barge	887.7	867.8	585	1252	22.83	0.7875

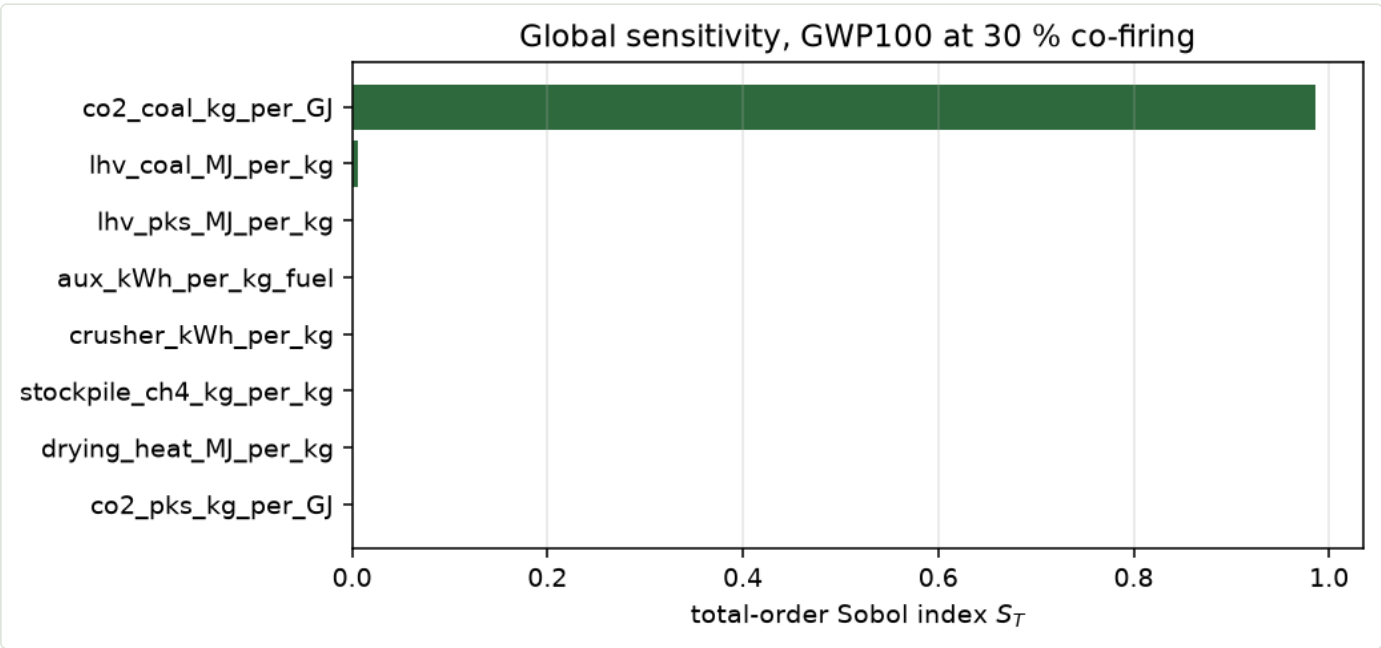
The decision-relevant number is the last column: the probability that a scenario beats the baseline once data quality is propagated. Low co-firing shares improve the mean while leaving the conclusion inside the noise; the benefit only becomes robust at the higher shares.

7. Sensitivity

Local elasticity ($\pm 10\%$, one at a time)

parameter	score_low	score_high	elasticity	abs_elasticity
co2_coal_kg_per_GJ	776.2	930	0.9014	0.9014
lhv_coal_MJ_per_kg	859.5	847.9	-0.06796	0.06796
lhv_pks_MJ_per_kg	855.5	851.2	-0.02485	0.02485
aux_kWh_per_kg_fuel	852.5	853.7	0.007182	0.007182
crusher_kWh_per_kg	852.8	853.4	0.003441	0.003441
stockpile_ch4_kg_per_kg	853	853.3	0.001422	0.001422
drying_heat_MJ_per_kg	853.1	853.2	0.0008558	0.0008558
co2_pks_kg_per_GJ	853.1	853.1	0	0

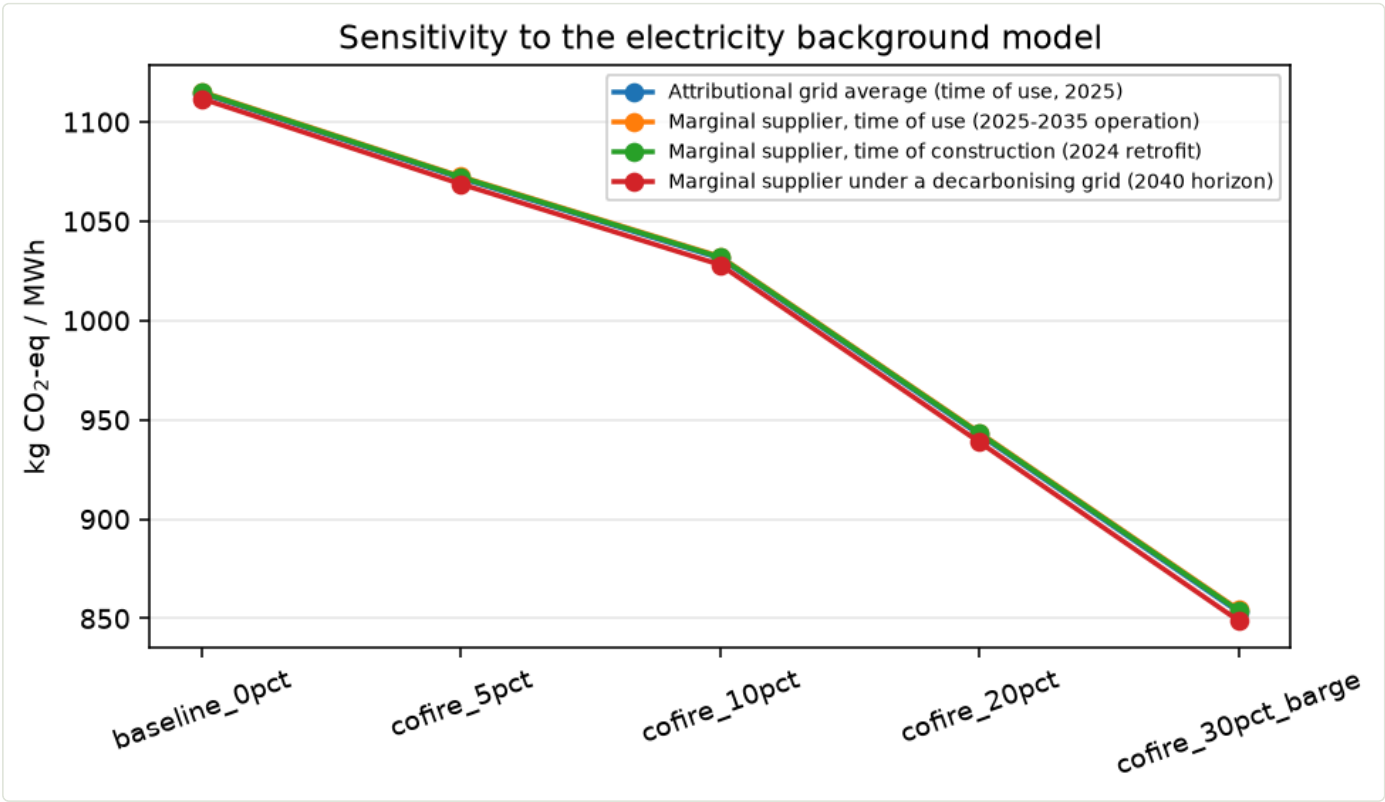
Global variance decomposition (Sobol)



parameter	S1	ST
co2_coal_kg_per_GJ	0.9808	0.986
lhv_coal_MJ_per_kg	0.005523	0.005833
lhv_pks_MJ_per_kg	0.01981	0.0007997
aux_kWh_per_kg_fuel	0.0003473	4.958e-05
crusher_kWh_per_kg	-0.0001805	1.212e-05
stockpile_ch4_kg_per_kg	-1.651e-05	2.287e-06
drying_heat_MJ_per_kg	0.0002239	8.969e-07
co2_pks_kg_per_GJ	0	0

8. Electricity model comparison

All scenarios re-run under each background assumption — time-of-use versus time-of-construction is a modelling choice with a quantified consequence, not a detail.



GWP100 (kg CO₂-eq / MWh)

	average_time_of_use	marginal_time_of_use	marginal_time_of_construction	marginal_decarbonising_2040
scenario				
baseline_0pct	1114	1115	1115	1112
cofire_5pct	1072	1073	1072	1069
cofire_10pct	1031	1032	1032	1028
cofire_20pct	942.5	943.6	943.2	938.7
cofire_30pct_barge	853.1	854.4	853.9	848.8

P(better than baseline)

	average_time_of_use	marginal_time_of_use	marginal_time_of_construction	marginal_decarbonising_2040
scenario				
baseline_0pct	0.000	0.000	0.000	0.000
cofire_5pct	0.520	0.520	0.520	0.520
cofire_10pct	0.578	0.578	0.578	0.580
cofire_20pct	0.670	0.670	0.670	0.672
cofire_30pct_barge	0.787	0.787	0.787	0.792

Verification

Every score is reproduced by an independent pandas matrix implementation ($g = CF \cdot Bs$); the largest relative difference in this run is 5.87e-08, i.e. float32 precision.

scenario	category	brightway	pandas_matrix	rel_diff
baseline_0pct	GWP100 (climate change)	1114.43	1114.43	3.28628e-09
baseline_0pct	Terrestrial acidification	3.32429	3.32429	4.31138e-09
baseline_0pct	Photochemical ozone formation	1.87649	1.87649	4.52798e-09
baseline_0pct	Fine particulate matter formation	2.15316	2.15316	5.87127e-08
baseline_0pct	Freshwater eutrophication	0.187247	0.187248	3.46609e-08
baseline_0pct	Land use	0	0	0
baseline_0pct	Water consumption	1.8	1.8	2.6491e-08
cofire_5pct	GWP100 (climate change)	1071.67	1071.67	2.03875e-08
cofire_5pct	Terrestrial acidification	3.22994	3.22994	4.69567e-08
cofire_5pct	Photochemical ozone formation	1.84844	1.84844	3.1186e-08
cofire_5pct	Fine particulate matter formation	2.10915	2.10915	1.43552e-08
cofire_5pct	Freshwater eutrophication	0.185841	0.185841	5.763e-08
cofire_5pct	Land use	13.5716	13.5716	1.48236e-09
cofire_5pct	Water consumption	1.81651	1.81651	8.42894e-09
cofire_10pct	GWP100 (climate change)	1031.29	1031.29	2.27047e-08
cofire_10pct	Terrestrial acidification	3.14355	3.14355	2.10829e-08
cofire_10pct	Photochemical ozone formation	1.8255	1.8255	1.26281e-08
cofire_10pct	Fine particulate matter formation	2.07072	2.07072	4.2147e-08
cofire_10pct	Freshwater eutrophication	0.18498	0.18498	3.15759e-08
cofire_10pct	Land use	27.4794	27.4794	1.01218e-08
cofire_10pct	Water consumption	1.83901	1.83901	1.60551e-09
cofire_20pct	GWP100 (climate change)	942.495	942.495	1.52566e-08
cofire_20pct	Terrestrial acidification	2.95134	2.95134	5.88038e-09
cofire_20pct	Photochemical ozone formation	1.77032	1.77032	2.03948e-08
cofire_20pct	Fine particulate matter formation	1.99037	1.99037	1.6938e-08
cofire_20pct	Freshwater eutrophication	0.183903	0.183903	2.77708e-08
cofire_20pct	Land use	55.999	55.999	1.74281e-08
cofire_20pct	Water consumption	1.87382	1.87382	2.74944e-08
cofire_30pct_barge	GWP100 (climate change)	853.137	853.137	1.93693e-08
cofire_30pct_barge	Terrestrial acidification	2.77146	2.77146	1.96735e-08
cofire_30pct_barge	Photochemical ozone formation	1.74948	1.74948	1.68049e-08
cofire_30pct_barge	Fine particulate matter formation	1.93823	1.93823	3.41255e-08
cofire_30pct_barge	Freshwater eutrophication	0.183708	0.183708	2.17253e-08
cofire_30pct_barge	Land use	85.8953	85.8953	6.97006e-09
cofire_30pct_barge	Water consumption	1.91613	1.91613	2.04703e-08

Proxy background factors stand in for licensed ecoinvent datasets; absolute values are indicative and the comparison between scenarios is the result that carries. Generated by `src/report.py`.